

POLAR BEAR

POLAR SCIENCE INTELLIGENCE & DISCOVERY PLATFORM

Comprehensive Operations Manual, Scientific Architecture Overview, Feature Catalog, and Practical Field Usage Guide for Arctic & Antarctic High-Latitude Research.

PLATFORM CLASSIFICATION

**Geospatial Intelligence & FAIR
Discovery**

SYSTEM BUILD

2026.4.19-LTS (Cryosphere Core)

COMPLIANCE STANDARDS

**W3C PROV-O · CF-1.8 NetCDF-4 · ISO
19115**

TARGET AUDIENCE

**Researchers, Data Managers,
Educators, Field Leads**

1. Executive Summary & Design System

1.1 The High-Latitude Research Challenge

Polar research is historically fragmented across disconnected oceanographic institutes, national Antarctic programs, satellite data nodes, and independent sensor deployments. Scientists seeking to understand sea-ice decline, ice-shelf calving, or ocean warming have traditionally had to manually correlate literature citations with heterogeneous NetCDF repositories, expedition drift logs, and isolated automated weather station feeds.

POLAR BEAR solves this fragmentation by providing a unified, source-grounded intelligence ecosystem that connects *Expeditions*, *Researchers*, *Datasets*, *Publications*, *Research Stations*, and *Live Sensor Observations* into a single topological research platform.

1.2 Core Architectural Principles & Zero-Gradient Philosophy

Unlike conventional SaaS applications, POLAR BEAR is designed as an institutional scientific instrument. The visual presentation adheres to strict guidelines to ensure maximum data density, zero visual noise, and absolute clarity:

- Solid Institutional Palette (Strictly Zero Gradients):** Deep Navy (#0B1F33), Dark Blue (#123B5D), Arctic Blue (#2F6F95), Ice White (#F7FAFC), White (#FFFFFF), Pale Ice (#E8F1F5), Dark Text (#16232E), Slate (#526474), and Border (#D8E1E7). No glowing or decorative gradients are permitted anywhere on the platform.
- Object-Oriented Cards Only:** Cards are strictly prohibited as generic decorative containers. Cards only represent discrete scientific entities (such as an expedition, dataset, paper, station, researcher, or observation). All general content is organized through structured tables, lists, tabs, timelines, and split-screen workspaces.
- FAIR Data Principles:** Every indexed dataset and publication adheres to Findable, Accessible, Interoperable, and Reusable standards, supported by persistent DataCite DOIs and W3C PROV-O cryptographic provenance tracking.

Institutional Design Standard Compliance

All typography uses a standardized Inter scale (H1: 36–42px, H2: 28px, H3: 20px, Body: 15px, Secondary: 14px, Metadata: 12px monospace). Navigation headers maintain a fixed 68px desktop height with strict `white-space: nowrap` preventing label wrapping across all viewports.

1.3 System Overview & Global Navigation Hierarchy

NAVIGATION SECTION	CLASSIFICATION	PRIMARY PURPOSE
Explore (Home)	Executive Discovery Workspace	9-stage homepage providing immediate intelligence discovery across all pillars.
Polar Map	Primary Geospatial Workbench	Interactive stereographic projection map for Arctic & Antarctic data exploration.
Research	Peer-Reviewed Literature	Indexed polar research papers with automated multi-language abstract translations.

NAVIGATION SECTION	CLASSIFICATION	PRIMARY PURPOSE
Data	Scientific Datasets Repository	Open-access NetCDF-4 time-series, grids, and mooring archives with DOI access.
Expeditions	Field Operations Tracker	Cruise tracks, icebreaker drift routes, expedition timelines, and scientific teams.
Stations	Observatories Directory	Permanent Arctic/Antarctic stations, atmospheric radars, and meteorological logs.
Education	Public Learning Gateway	Student Mode conceptual guides and teacher classroom curricula.
More / Secondary	Advanced Analysis & Governance	Knowledge Graph, Provenance, Live Telemetry, Community, Citizen Science, Jupyter.

2. Search, AI Intelligence & Geospatial Tools

Feature 1: Global Research Search & Federated Querying FEDERATED INDEX

WHAT IT IS

A unified, multi-repository search interface federating queries across publications, gridded datasets, expeditions, scientific stations, and researcher directories.

HOW IT WORKS

Combines dense vector semantic embeddings with sparse BM25 inverted indexing (Hybrid Mode) to allow researchers to query both precise keywords (e.g., "Bedmap3", "CTD Mooring") and conceptual natural language queries. Results are presented in an institutional tabular view with latency metrics.

HOW TO USE IT

- **Step 1:** Enter keywords into the search bar located on the homepage or via the top-right *Search* navbar button.
- **Step 2:** Use the **Search Mode** toggles (*Hybrid* , *Exact Keyword* , *Vector Semantic*) to refine query sensitivity.
- **Step 3:** Filter by **Discipline** (Sea Ice, Glaciology, Oceanography, Atmospheric) or **Region** (Antarctic, Arctic).
- **Step 4:** Click **Inspect** on any result row to immediately transition to the full entity profile.

Feature 2: Ask Polar Bear (Source-Grounded AI Research Assistant)

**ZERO-HALLUCINATION
PROTOCOL**

WHAT IT IS

An intelligent scientific assistant designed to synthesize complex polar phenomena exclusively from peer-reviewed literature, verified telemetry, and certified cruise logs.

HOW IT WORKS

Employs a strict Retrieval-Augmented Generation (RAG) architecture. When a user submits an inquiry, the system retrieves verified document chunks, computes cosine similarity over domain vector embeddings, extracts empirical evidence metrics (with confidence percentages), and produces a structured 3-part response: *Scientific Synthesis*, *Key Empirical Findings*, and *Cited Peer-Reviewed Sources* with verified DOIs.

HOW TO USE IT

- **Step 1:** Access from the Global Search bar ("Ask Polar Bear" button) or navigate to `/ask-ai` via the "More" dropdown.
- **Step 2:** Select from curated preset scientific inquiries or type a freeform polar science question.
- **Step 3:** Review the synthesized consensus text, empirical evidence metrics, and confidence intervals.
- **Step 4:** Click on linked citations to copy verified DOIs or jump directly to the underlying raw NetCDF dataset.

Feature 3: Polar Intelligence Map (Geospatial Workbench) NSIDC STEREOGRAPHIC

WHAT IT IS

An interactive, dual-hemisphere geospatial workbench providing high-latitude visualization of operational stations, vessel drift tracks, automated sensors, and oceanographic buoys.

HOW IT WORKS

Renders true polar projections using polar stereographic coordinate mathematics centered on 90°S (Antarctica) and 90°N (Arctic). Incorporates dynamic layer toggles for multi-sensor AMSR2 sea-ice extents, isothermal thermal fields, and sub-surface CTD moorings. Hovering dynamically computes geographic latitude and longitude coordinates.

HOW TO USE IT

- **Step 1:** Click **Polar Map** in the primary navbar.
- **Step 2:** Switch between **Antarctic (90°S)**, **Arctic (90°N)**, and **Global Corridor** projections using the top switcher.
- **Step 3:** Use the left **Map Layers & Telemetry** panel to toggle layers: Expeditions, Stations, Datasets, Sea Ice, Temperature, Ocean Obs, and Live Sensors.
- **Step 4:** Click any station marker (e.g., McMurdo, Concordia, Ny-Ålesund) to inspect real-time meteorological telemetry and active research projects.

3. Telemetry, Datasets & Field Operations

Feature 4: Live Polar Data Center & Anomaly Detection STREAMING INGESTION

WHAT IT IS

A real-time environmental monitoring center tracking Automated Weather Stations (AWS), drifting sea buoys, and moored CTD arrays across both polar caps.

HOW IT WORKS

Simulates continuous ingestion via WMO Global Telecommunication System (GTS) and Iridium satellite links. The system maintains two simultaneous data pipelines: **Raw Telemetry** (unfiltered data containing sensor spikes) and **Cleaned Data (QA/QC)** (where spike filters, median smoothing, and quality control flags have been applied). An integrated anomaly detection engine continuously compares current readings against a 30-year historical baseline ($+4.8\sigma$ envelope).

HOW TO USE IT

- **Step 1:** Access **Live Polar Data** via the homepage monitoring strip or under the "More" dropdown.
- **Step 2:** Toggle between **Cleaned Data (QA/QC)** and **Raw Telemetry** to compare raw sensor noise with validated readings.
- **Step 3:** Inspect the top **Anomaly Alert Banner** (e.g., Concordia AWS $+14.2^{\circ}\text{C}$ thermal surge detection).
- **Step 4:** Click on individual sensors in the sensor roster to inspect live time-series charts, standard deviations, and battery voltages.

Feature 5: Scientific Datasets Repository (CF-1.8 & NetCDF-4) OPEN ACCESS · DOI

WHAT IT IS

An open-access data repository hosting multidimensional climate grids, bed topography models, ice-core records, and aerosol observations.

HOW IT WORKS

All datasets are indexed according to Climate and Forecast (CF-1.8) conventions. Each entry includes spatial coordinate bounds (lat/lon), variable lists with standard physical units (m, $^{\circ}\text{C}$, kg/m^3), temporal resolution, license designations (CC-BY 4.0), and persistent DataCite DOIs. Pre-rendered spatial previews allow instant visual inspection of raster values.

HOW TO USE IT

- **Step 1:** Click **Data** in the primary navbar.
- **Step 2:** Select a dataset from the left list (e.g., *Bedmap3 Antarctic Bed Topography*, *Daily Sea-Ice Concentration*).
- **Step 3:** Navigate through the four standard tabs: **Overview** (metadata & citation), **Data** (tabular record view), **Visualization** (spatial grid preview), and **Provenance** (ingestion audit).
- **Step 4:** Use the **Copy DOI** button for publication references, or click **Download NetCDF** to export files.

Feature 6: Expeditions & Field Operations Tracking

OPERATIONS & LOGISTICS

WHAT IT IS

A comprehensive operational registry documenting polar research voyages, icebreaker drifts, deep field camps, and robotic sub-ice surveys.

HOW IT WORKS

Maintains structured expedition dossiers connecting vessel callsigns, voyage waypoints, chief scientists, participating international institutes, and direct links to all datasets and papers generated by the mission.

HOW TO USE IT

- **Step 1:** Click **Expeditions** in the navbar.
- **Step 2:** Select an expedition (e.g., *MOSAiC Arctic Drift*, *ITGC Thwaites Glacier*, *Weddell Sea Gyre*).
- **Step 3:** Explore the 8 structured tabs: **Overview**, **Route** (waypoint table), **Timeline** (milestones), **Team** (scientists & roles), **Datasets**, **Publications**, **Media**, and **Provenance**.

4. Provenance, Knowledge Graph & Analysis

Feature 7: FAIR Data Provenance & Cryptographic Lineage W3C PROV-O · SHA-256

WHAT IT IS

An auditable data lifecycle tracker certifying scientific reproducibility from initial sensor acquisition to final published journal paper.

HOW IT WORKS

Enforces a mandatory 6-stage lifecycle stepper: [Observation](#) → [Raw Dataset](#) → [Quality Check](#) → [Processed Dataset](#) → [Analysis](#) → [Publication](#). Every stage records exact timestamps, responsible agents, software versions, processing scripts, and immutable SHA-256 cryptographic hashes, enabling peer reviewers to verify that raw data has not been manipulated.

HOW TO USE IT

- **Step 1:** Select **Data Provenance** under the "More" navbar dropdown or from any dataset/expedition page.
- **Step 2:** Click through the 6 lifecycle stages to inspect processing transformations and QA algorithms.
- **Step 3:** Click **Verify SHA-256 Hash** to run an automated cryptographic integrity validation test.
- **Step 4:** Click **View Blockchain Receipt** to inspect immutable timestamped verification certificates.

Feature 8: Polar Science Knowledge Graph Explorer GRAPH TOPOLOGY

WHAT IT IS

An interactive topological graph mapping cross-disciplinary scientific relationships across researchers, expeditions, geography, datasets, and publications.

HOW IT WORKS

Builds a directed acyclic graph (DAG) where nodes represent discrete polar entities and edges represent relationships (e.g., "*Conducted By*", "*Deploys*", "*Derives*", "*Authored*"). Selecting any node dynamically computes and highlights connected relational sub-networks.

HOW TO USE IT

- **Step 1:** Access **Knowledge Graph** from the homepage or under the "More" dropdown.
- **Step 2:** Use the top filter buttons to isolate entity classes: *ALL*, *Researcher*, *Expedition*, *Location*, *Dataset*, or *Publication*.
- **Step 3:** Click any circular node on the SVG canvas to view its relational dossier and metadata in the right inspector panel.
- **Step 4:** Click **Inspect Full Record** to jump directly to the comprehensive view of that entity.

Feature 9: Collaborative Computational Workspace (Jupyter HPC)

REPRODUCIBLE PYTHON
CONTAINER

WHAT IT IS

A cloud-hosted Jupyter computational environment that co-locates open-source scientific code directly alongside high-latitude data repositories.

HOW IT WORKS

Connects to containerized compute clusters pre-configured with fundamental scientific Python libraries:

`xarray` (for multi-dimensional NetCDF grids), `cartopy` (for polar stereographic projections), `gsw` (TEOS-10 oceanographic equations), and `matplotlib`. Notebooks access datasets via local file descriptors without needing external data downloads.

HOW TO USE IT

- **Step 1:** Navigate to **Jupyter Workspace** under the "More" dropdown.
- **Step 2:** Browse the notebook catalog on the left (e.g., *amundsen_ice_thinning.ipynb*, *sea_ice_trend_amsr2.ipynb*).
- **Step 3:** Click **Run All Cells (HPC)** in the header to execute the notebook across the compute cluster.
- **Step 4:** Review real-time console stdout outputs, including calculated warming rates and provenance verification hashes.

5. Education, Community & Governance

Feature 10: Polar Science Education Portal **K-12 & UNIVERSITY CURRICULA**

WHAT IT IS

A dual-audience educational gateway translating complex cryospheric and climate phenomena for students and classroom instructors.

HOW IT WORKS

Provides two dedicated experiences: **Student Mode** presents simplified, jargon-free explanations of core concepts (e.g., Albedo Effect, Sea Ice vs. Icebergs, Ocean Conveyor Belts, Wildlife Adaptations). **Teacher Resources** provides peer-reviewed high school and undergraduate lab lesson plans incorporating authentic open-access datasets from Polar Bear archives.

HOW TO USE IT

- **Step 1:** Click **Education** in the main navbar.
- **Step 2:** Use the top mode switch to toggle between **Student Mode** and **Teacher Resources**.
- **Step 3:** In Student Mode, select topics on the left to read verified scientific summaries, key facts, and interactive tips.
- **Step 4:** In Teacher Resources, click **Download Lesson Materials (.ZIP)** to retrieve lab worksheets, Python notebook templates, and CSV datasets.

Feature 11: Citizen Science & Quarantined Staging Pool **QUARANTINED CONSENSUS**

WHAT IT IS

A public engagement framework allowing volunteers to classify satellite imagery, track wildlife colonies, and annotate ice features.

HOW IT WORKS

Maintains a strict **Scientific Separation Protocol**: Public annotations remain in a quarantined staging pool. Contributions only graduate into verified scientific repositories after achieving multi-user statistical consensus (>95% agreement) and receiving formal sign-off from credentialed institutional scientists.

HOW TO USE IT

- **Step 1:** Select **Citizen Science** under the "More" navbar dropdown.
- **Step 2:** Choose an active mission: *Weddell Seal Haul-Out Counting*, *Iceberg Calving Tracking*, or *Melt Pond Detection*.
- **Step 3:** Use interactive click-to-annotate tools to tag features on high-resolution satellite imagery.
- **Step 4:** Submit verified batches to earn researcher engagement points credited to your profile.

WHAT IT IS

Governance workspaces for peer review verification and infrastructural telemetry monitoring.

HOW IT WORKS

The **Review Workflow** guides submissions through a 4-stage quality gate: [Submitted](#) → [Under Review](#) → [Verified](#) → [Published](#) with FAIR audits checking metadata compliance, NetCDF standards, and DOI minting. The **Admin Dashboard** tracks server cluster health, SLURM compute node usage, Iridium satellite receiver uptime (99.99%), and storage utilization.

HOW TO USE IT

- **Step 1:** Select **Review Queue** or **Admin Dashboard** from the "More" menu or footer links.
- **Step 2:** In the review queue, inspect dataset submissions, audit compliance scores, and approve stage transitions.
- **Step 3:** In the admin console, review real-time ingestion rates (packets/sec) and cluster node loads.

6. Practical Workflows & Technical Reference

6.1 Typical User Workflows

Workflow A: Investigating Ocean Warming Beneath Ice Shelves

1. Open the **Global Search** and query `"Amundsen Sea basal melting"` or select *Glaciology* filter.
2. Use **Ask Polar Bear** to summarize recent findings on modified Circumpolar Deep Water (mCDW) intrusion.
3. Switch to the **Polar Intelligence Map**, select the *Antarctic (90°S)* projection, and activate the *Ocean Obs* and *Live Sensors* layers to inspect mooring temperatures.
4. Navigate to **Datasets** to download the verified NetCDF-4 time-series, copying the official DataCite DOI.
5. Inspect the **Data Provenance** tab to confirm the SHA-256 hash and W3C PROV-O audit trail.

Workflow B: Educational Curriculum Preparation

1. Navigate to the **Education Portal** and switch to **Teacher Resources**.
2. Select the curriculum unit: *"Antarctic Ice Shelves & Sea Level Rise"* (Grades 9–12).
3. Click **Download Lesson Materials (.ZIP)** to access student lab worksheets and real Bedmap3 CSV extracts.
4. Direct students to **Student Mode** to study the interactive explanations of sea ice dynamics and albedo feedback.

6.2 Standards & Technical Acronyms Reference

ACRONYM / STANDARD	FULL NAME	ROLE IN POLAR BEAR PLATFORM
CF-1.8	Climate & Forecast Metadata Convention	Standardizes NetCDF variable names, dimensions, and physical SI units for universal GIS interoperability.
W3C PROV-O	W3C Provenance Ontology	Formal data model describing the entities, activities, and agents involved in producing research artifacts.
FAIR	Findable, Accessible, Interoperable, Reusable	Guiding data stewardship principles enforced across all datasets and published literature.
AWS	Automated Weather Station	Autonomous polar meteorological station streaming temperature, wind, and pressure via satellite.
CTD	Conductivity, Temperature, Depth	Oceanographic sensor package measuring salinity and temperature profiles across the water column.
mCDW	Modified Circumpolar Deep Water	Dense, relatively warm (+0.4°C to +1.2°C) ocean water mass implicated in basal ice-shelf melting.
AMSR2	Advanced Microwave Scanning Radiometer 2	Satellite instrument measuring daily high-latitude sea-ice concentration and extent.

ACRONYM / STANDARD	FULL NAME	ROLE IN POLAR BEAR PLATFORM
PANGAEA	Data Publisher for Earth & Environmental Science	Federated archival partner repository integrated into POLAR BEAR federated search.

Support and Repository Inquiries

For data ingestion requests, WMO GTS telemetry integrations, or API keys, contact the Polar Bear Scientific Consortium at data@polarbear-science.org or visit the repository at <https://github.com/EdlaAbhishek/SIH.git>.